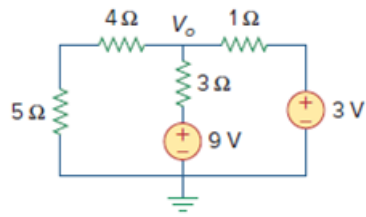


Problem

Using superposition, find V_o in the circuit of Fig. 4.76. Check with *PSpice* or *MultiSim*.

Figure. 4.76:



Step-by-step solution

Step 1 of 11

Consider the circuit as shown in Figure1.

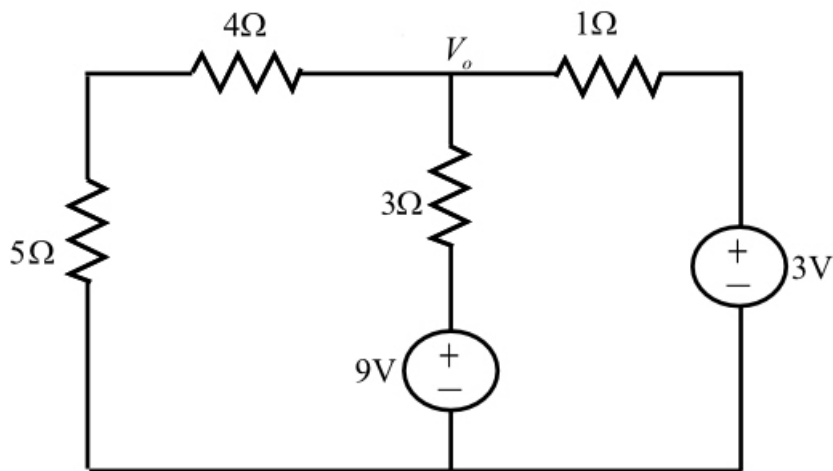


Figure 1

Step 2 of 11

To find the response V_o using superposition theorem, activate one independent source at a time and deactivate other independent sources present in the circuit. Find the individual response and add all the individual responses to get the final response.

The circuit contains 2 independent sources, so there will be three components of V_o .

$$V_o = V_o' + V_o''$$

Where,

V_o' is the response only due to 9V source.

V_o'' is the response only due to 3V source.

Step 3 of 11

To calculate the V_o' turn off the 3V voltage source that is replace the 3V voltage source with a short circuit as shown in Figure 2.

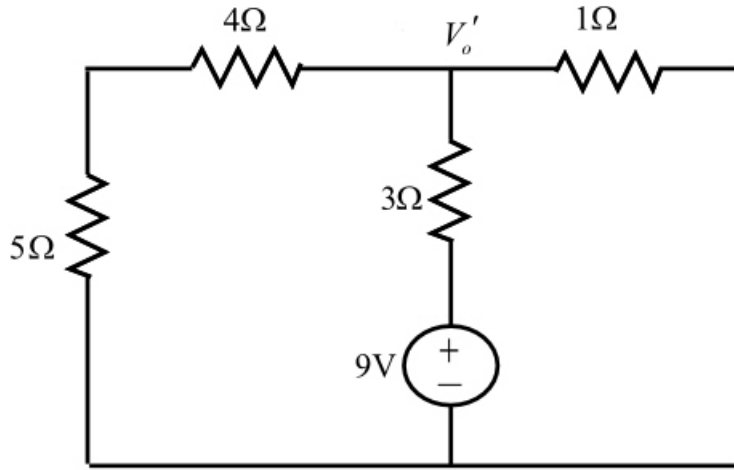


Figure 2

Step 4 of 11

Use the nodal analysis, at node V_o' .

$$\frac{V_o'}{(5+4)} + \frac{V_o' - 9}{3} + \frac{V_o'}{1} = 0$$

$$\frac{V_o'}{9} + \frac{V_o' - 9}{3} + \frac{V_o'}{1} = 0$$

$$\frac{V_o' + 3V_o' - 27 + 9V_o'}{9} = 0$$

$$13V_o' - 27 = 0$$

$$V_o' = 2.076 \text{ V}$$

Therefore, the value of V_o' is 2.076V.

Step 5 of 11

To calculate the V_o'' turn off the 9V voltage source that is replace the 9V voltage source with a short circuit as shown in Figure 3.

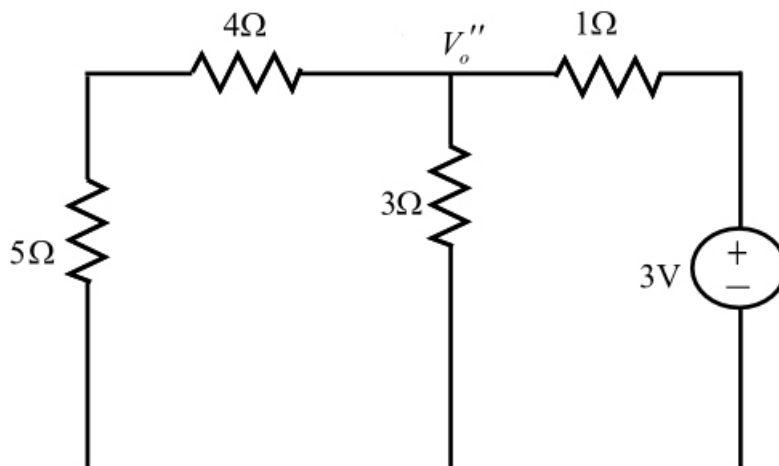


Figure 3

Step 6 of 11

Use the nodal analysis at node V_o'' .

$$\frac{V_o''}{(5+4)} + \frac{V_o''}{3} + \frac{V_o'' - 3}{1} = 0$$

$$\frac{V_o'}{9} + \frac{V_o'}{3} + \frac{V_o' - 3}{1} = 0$$

$$\frac{V_o' + 3V_o' + 9V_o' - 27}{9} = 0$$

$$13V_o' - 27 = 0$$

$$V_o' = 2.076 \text{ V}$$

Therefore, the value of V_o'' is 2.076V.

Step 7 of 11

Use superposition theorem to determine the value of voltage V_o .

$$V_o = V_o' + V_o''$$

Substitute 2.076 for V_o' , 2.076 for V_o'' , to obtain V_o .

$$\begin{aligned} V_o &= 2.076 + 2.076 \\ &= 4.152 \text{ V} \end{aligned}$$

Therefore, the value of V_o is 4.152 V.

Step 8 of 11

MultiSim Analysis:

- First invoke the MultiSim software.
- To open a new file, mouse click on 'File' and then schematic capture; it would open a working place.
- To draw a circuit, go to 'place' and 'components' for the components database.
- Select appropriate components and draw the circuit.
- Place a multimeter in parallel for voltage measurement.
- Then go to 'File' and 'save' the file.
- To run the simulation click on the 'run' icon.
- Then the multimeter shows the readings.

Step 9 of 11

Figure 4 shows the computer analysis using MultiSim.

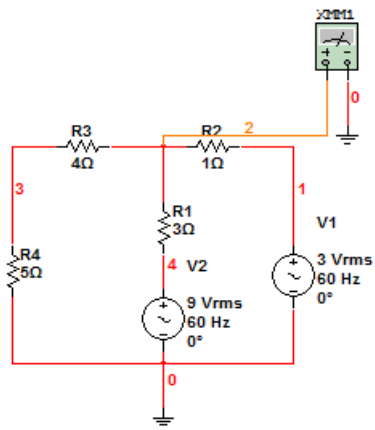


Figure 4

Step 10 of 11

The output of multimeter is shown in Figure 5.

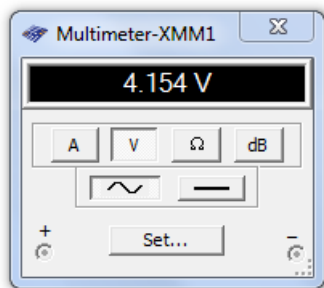


Figure 5

Step 11 of 11

We have an identical answer using the computer to analyze the circuit. We can see the voltage V_o multimeter reads 4.154 V and compares this to 4.15 V with the theoretical solution. A simplified analysis has produced the same result as a computer analysis as shown in Figure 5. Therefore, the results are verified using MultiSim.